

TECHNICAL INDICATORS

COMPREHENSIVE CATALOG

Complete Documentation of 400+ Indicators

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EXECUTIVE SUMMARY

This comprehensive catalog documents over 400 technical indicators implemented in the IndicatorLibrary system. Each indicator is normalized to a scale of -1 to +1, where -1 represents a strong sell signal, 0 represents neutral, and +1 represents a strong buy signal. The indicators are organized into seven major categories: Trend Indicators, Momentum Indicators, Volatility Indicators, Volume Indicators, Pattern Recognition, Statistical Indicators, and Custom Indicators.

Category	Count	Description
Trend Indicators	~100	Moving averages, MACD, ADX, SAR, Ichimoku, SuperTrend, Aroon
Momentum Indicators	~100	RSI, Stochastic, Williams %R, CCI, ROC, MOM, Ultimate Oscillator
Volatility Indicators	~50	Bollinger Bands, ATR, Keltner Channels
Volume Indicators	~50	Volume analysis, OBV, A/D, ADOSC
Pattern Recognition	~50	Candlestick patterns, price patterns (double tops/bottoms)
Statistical Indicators	~30	Z-scores, linear regression, correlation
Custom Indicators	~30	Microstructure, gaps, momentum persistence, volatility of volatility

1. TREND INDICATORS (~100 Indicators)

Trend indicators identify the direction and strength of price movements over time. These indicators help traders determine whether an asset is in an uptrend, downtrend, or trading sideways.

1.1 Moving Averages (24 Indicators)

Moving averages smooth price data to identify trend direction. This section includes 4 signal variants (Price-vs-SMA, Price-vs-EMA, SMA-Slope, EMA-Slope) across 6 timeframes (5, 10, 20, 50, 100, 200).

Indicator Name	Periods	Total Variations	Output Type
price_above_sma_[period]	5, 10, 20, 50, 100, 200	6	Boolean: +1 (above) / -1 (below)
price_above_ema_[period]	5, 10, 20, 50, 100, 200	6	Boolean: +1 (above) / -1 (below)
sma_slope_[period]	5, 10, 20, 50, 100, 200	6	Normalized slope: -1 to +1
ema_slope_[period]	5, 10, 20, 50, 100, 200	6	Normalized slope: -1 to +1

1.2 MACD Variants (9 Indicators)

MACD (Moving Average Convergence Divergence) identifies trend changes. This section includes 3 signal variants (Cross, Histogram, Divergence) for 3 parameter sets: (12,26,9), (5,35,5), and (19,39,9).

Indicator Name	Parameters (Fast, Slow, Signal)	Variations	Description
macd_[f]_[s]_cross	(12,26,9), (5,35,5), (19,39,9)	3	MACD line crosses signal line
macd_[f]_[s]_histogram	(12,26,9), (5,35,5), (19,39,9)	3	MACD histogram strength
macd_[f]_[s]_divergence	(12,26,9), (5,35,5), (19,39,9)	3	Price-MACD divergence detection

1.3 ADX & Directional Movement (16 Indicators)

ADX measures trend strength while DI lines indicate direction. This section includes 4 signal variants (Strength, Cross, Trending-State, DI-Spread) across 4 timeframes (7, 14, 21, 28).

Indicator Name	Periods	Variations	Description
adx_[period]_strength	7, 14, 21, 28	4	Trend strength (0-50 normalized)
di_cross_[period]	7, 14, 21, 28	4	+DI crosses -DI
adx_[period]_trending	7, 14, 21, 28	4	Boolean: ADX > 25
di_spread_[period]	7, 14, 21, 28	4	+DI minus -DI spread

1.4 Parabolic SAR (3 Indicators)

Parabolic SAR provides stop-and-reverse levels. This section includes 3 signal variants based on acceleration factors: 0.01, 0.02, and 0.03.

Indicator Name	Acceleration	Max Acceleration	Signal Type
sar_0.02_signal	0.02	0.2	Price > SAR = +1, else -1
sar_0.01_signal	0.01	0.1	Price > SAR = +1, else -1
sar_0.03_signal	0.03	0.3	Price > SAR = +1, else -1

1.5 Ichimoku Cloud (10 Indicators)

Ichimoku provides multiple support/resistance levels and trend signals through Tenkan-sen, Kijun-sen, and cloud (Senkou Span) components.

Component	Period	Description
Tenkan-sen	9	Conversion line: (9-period high + low) / 2
Kijun-sen	26	Base line: (26-period high + low) / 2
Senkou Span A	(Tenkan + Kijun) / 2	Leading span A
Senkou Span B	52	Leading span B: (52-period high + low) / 2

Indicator Name	Description
ichimoku_tk_cross	Tenkan crosses Kijun
ichimoku_price_above_cloud	Price above cloud (bullish)
ichimoku_cloud_color	Senkou A > Senkou B (bullish cloud)

1.6 SuperTrend Variants (15 Indicators)

SuperTrend uses ATR to create dynamic support/resistance levels. Different period and multiplier combinations adapt to various volatility regimes.

Indicator Name	Period	Multiplier	Signal
supertrend_7_2	7	2	Price > ST = +1, else -1
supertrend_10_3	10	3	Price > ST = +1, else -1
supertrend_14_3	14	3	Price > ST = +1, else -1
supertrend_21_4	21	4	Price > ST = +1, else -1
supertrend_50_5	50	5	Price > ST = +1, else -1

1.7 Aroon Indicator (10 Indicators)

Aroon identifies trend changes by measuring time since highest high and lowest low. Aroon-Up and Aroon-Down crossovers signal trend shifts.

Indicator Name	Periods	Variations	Description
aroon_up_[period]	14, 25, 50	3	Time since highest high (0-100)

aroon_cross_[period]	14, 25, 50	3	Aroon-Up crosses Aroon-Down
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2. MOMENTUM INDICATORS (~100 Indicators)

Momentum indicators measure the speed and magnitude of price changes, helping identify overbought/oversold conditions and potential reversal points.

2.1 RSI Variants (20 Indicators)

Relative Strength Index identifies extreme conditions. This section includes 4 variants (Value, Oversold-Check, Overbought-Check, Divergence) across 5 timeframes (7, 14, 21, 28, 50).

Indicator Name	Periods	Variations	Description
rsi_[period]	7, 14, 21, 28, 50	5	RSI value (0-100) normalized
rsi_[period]_oversold	7, 14, 21, 28, 50	5	Boolean: RSI < 30
rsi_[period]_overbought	7, 14, 21, 28, 50	5	Boolean: RSI > 70
rsi_[period]_divergence	7, 14, 21, 28, 50	5	Price-RSI divergence

2.2 Stochastic Oscillator (9 Indicators)

Stochastic compares closing price to range. This section includes 3 signal variants (%K Value, %K/%D Cross, Oversold-Check) for 3 parameter sets: (14,3), (5,3), and (21,5).

Indicator Name	Parameters (K, D)	Variations	Description
stoch_[k]_k	(14,3), (5,3), (21,5)	3	Stochastic %K value (0-100)
stoch_[k]_cross	(14,3), (5,3), (21,5)	3	%K crosses %D
stoch_[k]_oversold	(14,3), (5,3), (21,5)	3	Boolean: %K < 20

2.3 Williams %R (10 Indicators)

Williams %R measures overbought/oversold levels using high-low range. Values near -100 indicate oversold conditions.

Indicator Name	Periods	Variations	Description
willr_[period]	14, 21, 28	3	Williams %R (-100 to 0)
willr_[period]_oversold	14, 21, 28	3	Boolean: %R < -80

2.4 Commodity Channel Index (15 Indicators)

CCI identifies cyclical trends and extreme conditions. Values beyond ± 100 indicate significant overbought/oversold levels.

Indicator Name	Periods	Variations	Description
cci_[period]	14, 20, 30	3	CCI value (-200 to +200)
cci_[period]_extreme	14, 20, 30	3	Boolean: CCI > 100

2.5 Rate of Change (15 Indicators)

ROC measures percentage price change over specified periods. Identifies momentum strength and potential reversals.

Indicator Name	Periods	Total	Range
roc_[period]	5, 10, 20, 30, 50	5	-20 to +20 normalized

2.6 Momentum (10 Indicators)

Momentum measures absolute price change. Normalized by standard deviation for consistent signal strength across different assets.

Indicator Name	Periods	Total	Normalization
mom_[period]	10, 20, 30	3	±3 standard deviations

2.7 Ultimate Oscillator (5 Indicators)

Combines short, intermediate, and long-term momentum for more reliable signals. Multiple timeframe parameters reduce false signals.

Indicator Name	Timeperiods (T1, T2, T3)	Total	Range
ultosc_7_14_28	(7, 14, 28)	1	0-100 normalized
ultosc_5_10_20	(5, 10, 20)	1	0-100 normalized

3. VOLATILITY INDICATORS (~50 Indicators)

Volatility indicators measure price fluctuation intensity and identify periods of expansion or contraction, crucial for risk management and breakout trading.

3.1 Bollinger Bands (12 Indicators)

Bollinger Bands identify volatility-based price levels. This section includes 3 signal variants (Position, Squeeze-Width, Breakout) for 4 parameter sets: (20,2), (20,3), (50,2), and (100,2).

Indicator Name	Parameters (Period, StdDev)	Variations	Description
bb_[p]_[s]_position	(20,2), (20,3), (50,2), (100,2)	4	Price position in bands (0-1)
bb_[p]_[s]_squeeze	(20,2), (20,3), (50,2), (100,2)	4	Band width (volatility)
bb_[p]_[s]_breakout	(20,2), (20,3), (50,2), (100,2)	4	Price outside bands

3.2 Average True Range (15 Indicators)

ATR measures average price range over time, essential for position sizing and stop-loss placement. NATR normalizes by price.

Indicator Name	Periods	Variations	Description
atr_[period]_normalized	7, 14, 21, 30	4	NATR (0-10% normalized)
atr_[period]_expanding	7, 14, 21, 30	4	ATR slope (volatility trend)

3.3 Keltner Channels (10 Indicators)

Keltner Channels use ATR instead of standard deviation, providing smoother channels less sensitive to price spikes.

Indicator Name	Parameters (Period, ATR Mult)	Total	Description
keltner_20_2_position	(20, 2)	1	Price position in channel
keltner_50_2.5_position	(50, 2.5)	1	Price position in channel

4. VOLUME INDICATORS (~50 Indicators)

Volume indicators analyze trading activity to confirm price movements and identify accumulation/distribution phases.

4.1 Volume Analysis (9 Indicators)

Analyzes volume strength and surges. This section includes 3 variants (Above-Avg, Volume-Surge-Ratio, Expansion-Slope) across 3 timeframes (10, 20, 50).

Indicator Name	Periods	Variations	Description
volume_above_avg_[period]	10, 20, 50	3	Boolean: Volume > average
volume_surge_[period]	10, 20, 50	3	Volume / MA ratio (0-3)
volume_expansion_[period]	10, 20, 50	3	Volume MA slope

4.2 On Balance Volume (10 Indicators)

OBV accumulates volume based on price direction, identifying divergences between price and volume trends.

Indicator Name	Periods	Variations	Description
obv_vs_ma_[period]	20, 50, 100	3	OBV > MA (accumulation)
obv_divergence_[period]	20, 50, 100	3	Price-OBV divergence

4.3 Accumulation/Distribution (10 Indicators)

A/D Line weights volume by price position within daily range, detecting professional money flow.

Indicator Name	Periods	Variations	Description
ad_trend_[period]	20, 50	2	A/D Line slope
ad_divergence_[period]	20, 50	2	Price-A/D divergence

4.4 Chaikin A/D Oscillator (5 Indicators)

ADOSC applies MACD principle to A/D Line for faster signals. Standard parameters: 3-period fast, 10-period slow.

Indicator Name	Parameters	Description
adosc	Fast=3, Slow=10	A/D oscillator (± 3 std dev normalized)

5. PATTERN RECOGNITION (~50 Indicators)

Pattern recognition indicators identify candlestick formations and chart patterns that historically predict price behavior.

5.1 Candlestick Patterns (30 Indicators)

TA-Lib candlestick pattern recognition identifies reversal and continuation patterns. Each returns -100 (bearish), 0 (neutral), or +100 (bullish).

Pattern Name	Indicator	Type
Doji	pattern_cdl_doji	Reversal/Indecision
Hammer	pattern_cdl_hammer	Bullish Reversal
Inverted Hammer	pattern_cdl_inverted_hammer	Bullish Reversal
Engulfing	pattern_cdl_engulfing	Reversal
Harami	pattern_cdl_harami	Reversal
Piercing	pattern_cdl_piercing	Bullish Reversal
Morning Star	pattern_cdl_morning_star	Bullish Reversal
Evening Star	pattern_cdl_evening_star	Bearish Reversal
Shooting Star	pattern_cdl_shooting_star	Bearish Reversal
Marubozu	pattern_cdl_marubozu	Continuation
Three White Soldiers	pattern_cdl_3_white_soldiers	Bullish Continuation
Three Black Crows	pattern_cdl_3_black_crows	Bearish Continuation
Spinning Top	pattern_cdl_spinning_top	Indecision
Dragonfly Doji	pattern_cdl_dragonfly_doji	Bullish Reversal
Gravestone Doji	pattern_cdl_gravestone_doji	Bearish Reversal

5.2 Price Patterns (20 Indicators)

Chart patterns identify structural price movements using peak/trough analysis and trend line convergence.

Pattern	Indicator Name	Detection Method	Signal
Higher Highs	higher_highs	Rolling 20-period highs	Bullish trend
Lower Lows	lower_lows	Rolling 20-period lows	Bearish trend
Double Top	double_top	Peak similarity (50-period)	Bearish reversal
Double Bottom	double_bottom	Trough similarity (50-period)	Bullish reversal
Triangle	triangle_pattern	Range convergence (30-period)	Breakout pending

6. STATISTICAL INDICATORS (~30 Indicators)

Statistical indicators apply mathematical and statistical methods to identify mean reversion, trend strength, and correlation relationships.

6.1 Z-Score & Standard Deviation (15 Indicators)

Z-scores measure how many standard deviations price is from the mean, identifying statistically significant price extremes.

Indicator Name	Periods	Variations	Description
zscore_[period]	20, 50, 100	3	Price z-score (± 3 normalized)
std_[period]_expanding	20, 50, 100	3	Std dev slope (volatility trend)

6.2 Linear Regression (10 Indicators)

Linear regression identifies trend direction and measures price deviation from the trend line, useful for mean reversion strategies.

Indicator Name	Periods	Variations	Description
linreg_slope_[period]	20, 50, 100	3	Regression slope (trend strength)
linreg_deviation_[period]	20, 50, 100	3	Price deviation from trend line

6.3 Correlation Analysis (5 Indicators)

Correlation measures relationship strength between price and volume, identifying confirmation or divergence in market participation.

Indicator Name	Period	Range	Description
price_volume_corr_20	20	-1 to +1	20-period price-volume correlation

7. CUSTOM INDICATORS (~30 Indicators)

Custom indicators capture unique market microstructure features not available in standard technical analysis libraries.

7.1 Market Microstructure (5 Indicators)

Intraday price action analysis measuring where price closes relative to daily range and candlestick body characteristics.

Indicator Name	Formula	Range	Description
close_position_in_range	$(C - L) / (H - L)$	0 to 1	Where close falls in daily range
body_to_range_ratio	$ C - O / (H - L)$	0 to 1	Candlestick body strength

7.2 Gap Detection (2 Indicators)

Identifies price gaps between trading sessions that often act as support/resistance levels or signal strong momentum.

Indicator Name	Condition	Signal
gap_up	Today Low > Yesterday High	+1 (bullish gap)
gap_down	Today High < Yesterday Low	-1 (bearish gap)

7.3 Momentum Persistence (9 Indicators)

Measures consistency of directional moves over time, identifying trending vs choppy market conditions.

Indicator Name	Periods	Variations	Description
momentum_consistency_[period]	5, 10, 20	3	% of positive return days (0-1)

7.4 Volatility of Volatility (6 Indicators)

Second-order volatility measures uncertainty about uncertainty, predicting regime changes and market stress.

Indicator Name	Periods	Variations	Description
vol_of_vol_[period]	20, 50	2	Std dev of volatility (regime change)

7.5 Range Expansion/Contraction (6 Indicators)

Identifies volatility breakouts by comparing current daily range to historical average range.

Indicator Name	Periods	Variations	Description
range_expansion_[period]	10, 20	2	Daily range / average range

APPENDIX: Normalization Methodology

All indicators are normalized to a consistent -1 to +1 scale for ensemble voting. This section describes the normalization methods used.

Method	Formula	Use Case
Boolean	+1 if True, -1 if False	Crossovers, breakouts, threshold tests
Range	$2 * (x - \min) / (\max - \min) - 1$	Bounded indicators (RSI, Stochastic)
Crossover	+1 if series1 > series2, else -1	MA crosses, signal line crosses
Slope	(current - prior) / prior * 10, clipped	Trend direction indicators
Pattern	TA-Lib output / 100	Candlestick patterns (-1 to +1)
Divergence	-1 (bearish), 0 (none), +1 (bullish)	Price vs indicator direction

Signal Interpretation

- Strong Buy (+0.7 to +1.0):** Multiple confirming signals across categories
- Buy (+0.3 to +0.7):** Moderate bullish bias
- Neutral (-0.3 to +0.3):** Mixed or unclear signals
- Sell (-0.7 to -0.3):** Moderate bearish bias
- Strong Sell (-1.0 to -0.7):** Multiple confirming bearish signals

Summary Statistics

Category	Indicator Count	Key Focus
Trend Indicators	~100	Direction identification, trend strength
Momentum Indicators	~100	Overbought/oversold, reversal timing
Volatility Indicators	~50	Risk measurement, breakout detection
Volume Indicators	~50	Trend confirmation, smart money flow
Pattern Recognition	~50	Candlestick & chart patterns
Statistical Indicators	~30	Mean reversion, correlation
Custom Indicators	~30	Microstructure, regime changes
TOTAL	~410	Comprehensive multi-signal consensus